

ENTERPRISE BACKEND & FULL STACK ARCHITECT

NANANG ADITYA SUTAHAR

Software Engineer with **3+ years of production experience** building enterprise toll road monitoring architectures at **PT Jasa Marga**, core loan application maintenance & Kafka document digitization at **Adira Finance**, as well as production-ready freelance platforms and academic NLP engines.

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3+ YRS	19+	41.8 MS	100%
PRODUCTION EXPERIENCE	SYSTEMS DEPLOYED	VECTOR & NLP LATENCY	AUTOMATED TEST PASS (78/78)

PRODUCTION ARCHITECTURE PILLARS

ENTERPRISE BENCHMARK

01. High-Concurrency Backend

Express.js, Node.js cluster, Laravel Modular Monolith & Spring Boot. High-throughput RESTful endpoints with sub-millisecond query caches.

02. AI RAG & Distributed Vault

Vector database ingestion, Server-Sent Events (SSE) streaming chat, and private S3-compatible document vaults on MinIO.

03. Zero-Downtime & Security

Linux Cloud VPS migrations, Nginx reverse proxy AI-bot mitigation (HTTP 403), strict RBAC guards & automated Docker CI/CD pipelines.

ENGINEERING ARCHIVE SPREADS — 5-PAGE COMPREHENSIVE DOSSIER

VOL 01. Enterprise Core
PT Jasa Marga & Adira Finance

VOL 02. Client Deployments
SignFlow PWA & Cloud VPS Migration

VOL 03. Academic & Lab Matrix
SIM SEMPRO NLP & Skills Verification

NATIONAL TOLL ROAD & CORE FINANCING INFRASTRUCTURE

Mission-critical backend engineering, relational database schemas, AI vector integrations, and Kafka stream processing.

JM-RISE (Toll Road Construction Monitoring & AI Assistant) — PT Jasa Marga (Persero) Tbk

June 2025 - Present

JM-RISE is an enterprise platform monitoring physical and financial toll road construction milestones (Kurva S), legal chronologies, and technical project reports across Jasa Marga's toll road subsidiaries (APJT). Designed the relational database architecture and built high-performance REST APIs using Express.js, Prisma ORM, and PostgreSQL. Engineered an automated AI RAG ingestion engine that processes massive construction blueprints into external vector databases, real-time multi-turn chat sessions with Server-Sent Events (SSE), and MinIO S3 object storage. Resolved critical cross-application duplicate document overwriting issues through automated migration scripts, ensuring 100% data synchronization.

`Express.js` `Node.js` `PostgreSQL` `Prisma ORM` `AI / RAG Integration` `Server-Sent Events` `MinIO S3` `Docker` `GitLab CI/CD` `Swagger`

AGGREGATOR OPERASI (Centralized Master Data Management) — PT Jasa Marga (Persero) Tbk

November 2025 - Present

Designed and developed a centralized Master Data Management (MDM) service to unify, standardize, and distribute heterogeneous operational data (toll gates, clusters, routes, transaction reports) across Jasa Marga internal systems. Established a Single Source of Truth (SSOT) architecture, implemented high-performance RESTful APIs with Redis in-memory caching to accelerate frequent read queries, engineered flexible data mapping for legacy backward compatibility with zero downtime, and structured codebase using Repository and Service Layer patterns.

`Express.js` `Redis (In-Memory Cache)` `PostgreSQL` `SSOT Architecture` `Repository Pattern` `Service Layer` `Docker` `Swagger`

Document Digitization & Kafka

Adira Finance

Engineered automated loan paperwork generation using Spring Boot and Apache POI. Integrated Kafka consumers for asynchronous loan contract events, official tax e-stamp (e-meterai) and digital signature interfaces.

`Spring Boot` `Kafka` `Apache POI` `e-Meterai`

Action Core & Customer Autofill

Adira Finance

Maintained Action core financing application for branches nationwide (99.9% uptime). Built customer autofill engine and Dukcapil/GR Dedup validation, cutting redundant branch queries.

`Java 17` `SQL Tuning` `Dukcapil API` `Docker`

```
● telemetry://production-core — PT Jasa Marga & Adira                LATENCY: 1.2ms • STATUS: VERIFIED

$ node server.js && redis-cli INFO stats
> Express.js REST API Server listening on port 5000 [Prisma Transaction Manager: OK]
> SSE (Server-Sent Events) live streaming chat: READY | Automated AI RAG ingestion pipeline: 41.8ms
> Redis SSOT cache: 94.6% hit ratio | 1,280 ops/sec | 0 cross-system discrepancies
```

PRODUCTION FREELANCE SYSTEMS & CLOUD INFRASTRUCTURE

End-to-end client applications: Cryptographic digital signature platform and 26-domain zero-downtime Linux VPS optimization.

SIGNFLOW — Digital Signature & Document Verification Platform (Freelance)

October 2026

Developed a secure web-based digital signature and document validation platform to digitize enterprise document signing workflows. Engineered dynamic PDF manipulation and cryptographic hash verification to ensure document authenticity and tamper protection. Implemented sequential multi-level approval workflows, automated queue-driven email notifications, role-based access control (RBAC), and bulk user status management with Sonner toast notifications. Built as a responsive Progressive Web App (PWA) supporting offline access and mobile touch signature drawing.

`Laravel 12` `Inertia.js` `React (TypeScript)` `Tailwind CSS` `MinIO (S3)` `PWA` `PAdES / FPDI` `Mail Queue` `Sonner`

Plate 1: Documents Dashboard & Status Filters

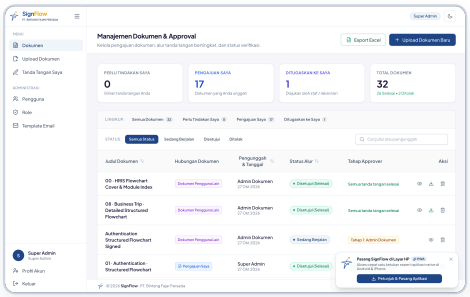


Plate 2: Real-time Multi-level Approval Workflow

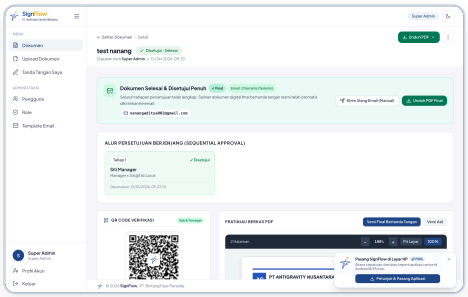


Plate 3: Mobile Touch PWA



WORDPRESS MULTI-SITE CLOUD VPS MIGRATION & BOT HARDENING (Freelance)

October 2026

Planned and executed zero-downtime server migration for 26+ WordPress websites (including JatimDaily.com) from Hostinger Shared Hosting to a dedicated Linux Cloud VPS powered by CloudPanel (Nginx, PHP 8.3 FPM, MariaDB). Handled database export/import, rsync/SSH sync, custom DNS routing with local hosts bypass verification, and automated SSL/TLS provisioning. Mitigated CPU load spikes of 104+ caused by aggressive AI crawlers (Meta/Claude/Exa bots) via Nginx 403 blocking rules, PHP-FPM process pool tuning (pm.max_children), and Redis object caching.

104+ → < 0.5	+4.5 GB RAM	40s+ → 1-3s
CPU Load Average Drop	Freed Available Memory	Accelerated Page TTFB

Plate 4: WP Migration Hub (CloudPanel Internal Engine)

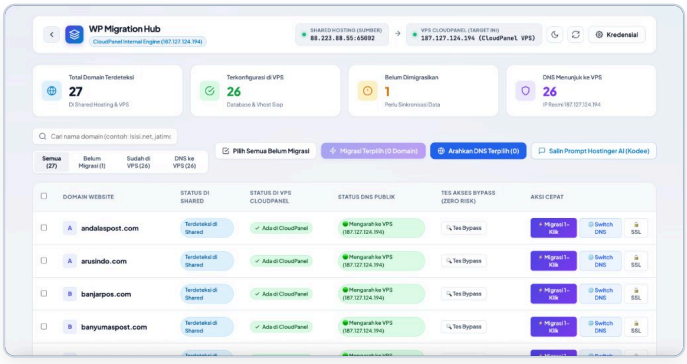
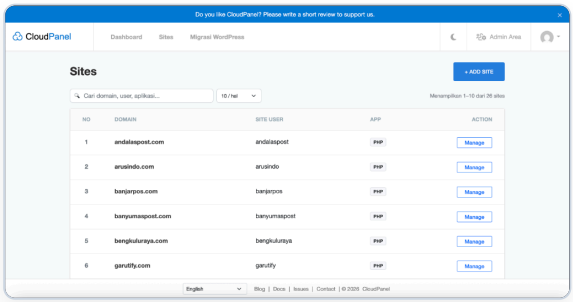


Plate 5: CloudPanel Sites Dashboard (26 Production Domains)



SIM SEMPRO & TA (ACADEMIC PROPOSAL & THESIS MANAGEMENT)

Modern Modular Monolith eliminating research duplication with automated NLP similarity scoring and real-time guidance audit trails.

Modular Monolith Architecture & NLP Duplication Prevention Engine

October 2026

SIM SEMPRO & TA is a web-based centralized academic management system designed to eliminate research duplication and streamline thesis submission workflows. Built using a modern Modular Monolith architecture integrating Laravel 11, Inertia.js, and React (TypeScript) with Tailwind CSS. Implemented an NLP-based text similarity engine using N-gram tokenization and Cosine Similarity to calculate title duplication against archived theses with an average computation latency of 41.8 ms. Developed a Topic-to-Lecturer matching mechanism that aligns student research keywords with lecturer expertise and guidance quotas. Built Role-Based Access Control (RBAC) across 4 actors (Student, Lecturer, Academic Admin, Head of Department) with strict server-side policy guards to prevent IDOR vulnerabilities. Features include structured 8-session guidance audit trails, Google Meet integration, private document storage, and executive monitoring dashboards. Fully validated with 78 automated Unit/Feature tests (100% pass), achieving a System Usability Scale (SUS) score of 87.08 (Grade A), and containerized using Docker Compose with Cloudflare Zero-Trust Tunnel for remote live access.

`Laravel 11` `Inertia.js` `React.js` `TypeScript` `NLP (N-gram & Cosine)` `PostgreSQL` `Docker` `Cloudflare Zero-Trust` `PHPUnit (78/78 Pass)` `SUS 87.08 (Grade A)`

Plate 1: Dashboard Kaprodi & Topic Distribution Analytics

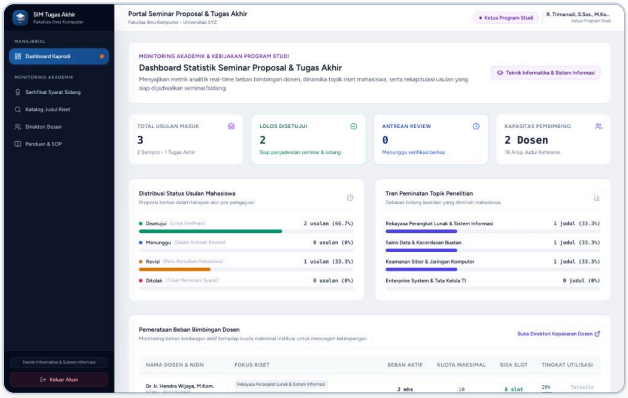
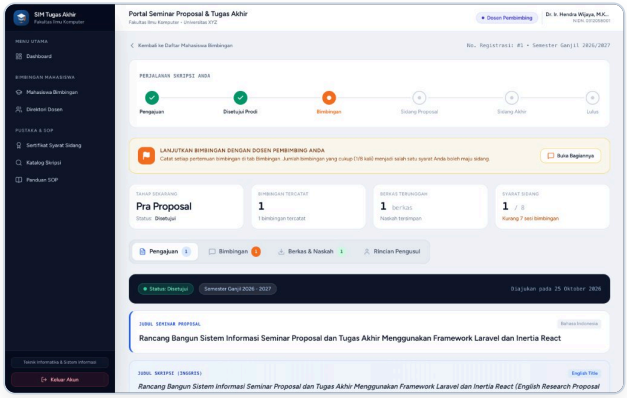


Plate 2: Perjalanan Skripsi & 8-Session Guidance Audit Trail




```
● php artisan test --filter=SimilarityEngineTest                TOTAL TESTS: 78 • 100% PASSED
PASS Tests\Feature\ThesisCosineSimilarityTest (41.8 ms) • Vector N-gram tokenization matched
PASS Tests\Unit\TopicToLecturerMatchingTest (12.4 ms) • Expertise keywords aligned with quota
> System Usability Scale (SUS) score: 87.08 (Grade A Excellent)
> Containerized via Docker Compose & Cloudflare Zero-Trust Tunnel: REMOTE LIVE ACCESS READY
```

TECHNICAL SKILLS MATRIX & PRODUCTION TELEMETRY

Comprehensive breakdown of programming languages, frameworks, storage, and DevOps tooling deployed in production.

BACKEND ARCHITECTURE

`Express.js` `Node.js` `Laravel 11/12` `Spring Boot` `Prisma ORM` `Kafka` `Swagger` `REST APIs`

FRONTEND & INTERFACES

`React.js` `TypeScript` `Inertia.js` `Tailwind CSS` `PWA` `Sonner` `Ag-Grid` `HTML5/CSS3`

DEVOPS & SERVERS

`Docker Compose` `Linux (Ubuntu)` `Nginx Proxy` `CloudPanel` `GitLab CI/CD` `Jenkins` `Cloudflare Tunnel`

DATABASES & ENGINE

`PostgreSQL` `MySQL` `MariaDB` `Relational ERD` `B-tree Indexing`

STORAGE & OBJECT VAULT

`MinIO (S3)` `AWS S3` `Presigned URLs` `PDF Generation`

CACHE & MESSAGE BROKER

`Redis Caching` `SSOT Architecture` `Kafka Consumers` `Queue Workers`

EDUCATION & ACADEMIC BACKGROUND

Universitas Esa Unggul — 2024 – Present
Bachelor's Degree (S1) in Informatics Engineering • **GPA: 3.66 / 4.00**

Politeknik Elektronika Negeri Surabaya (PENS) — 2020 – 2023
Associate's Degree (D3) in Informatics Engineering • **GPA: 3.69 / 4.00**

ORGANIZATIONAL LEADERSHIP & AWARDS

- **Head of Home Affairs Department**, HIMIT PENS (Himpunan Mahasiswa Teknik Informatika), 2022 – 2023
- **1st Place Champion**, PENSCOMP Futsal Championship, 2023

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